

Hadoop Distributed File System

Agenda → Installation Survey ✓
→ Internals of HDFS ✓
→ Play with HDFS ✓

Recap →

Data Sources

Realtime & Batch

Ingestion

Data Lake & Data warehouse

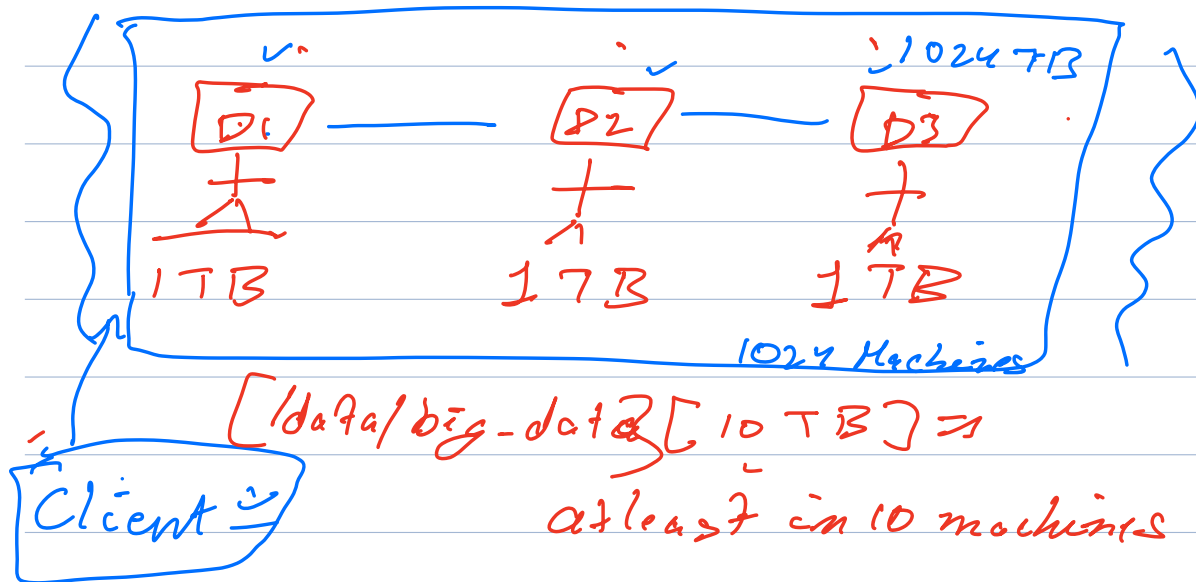
ETL

Airflow

Reporting

HDFS → Hadoop Data File System.

- ① Amount of data is large
 - ② Analytical data
- } Need of Big Data



Yahoo - HDFS 1.0 -> Solution

① Software which maintains index of data

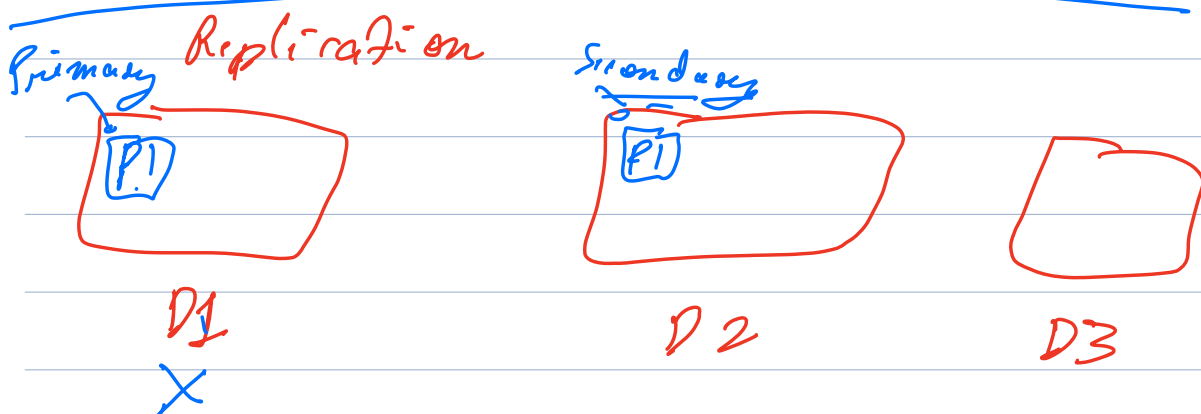
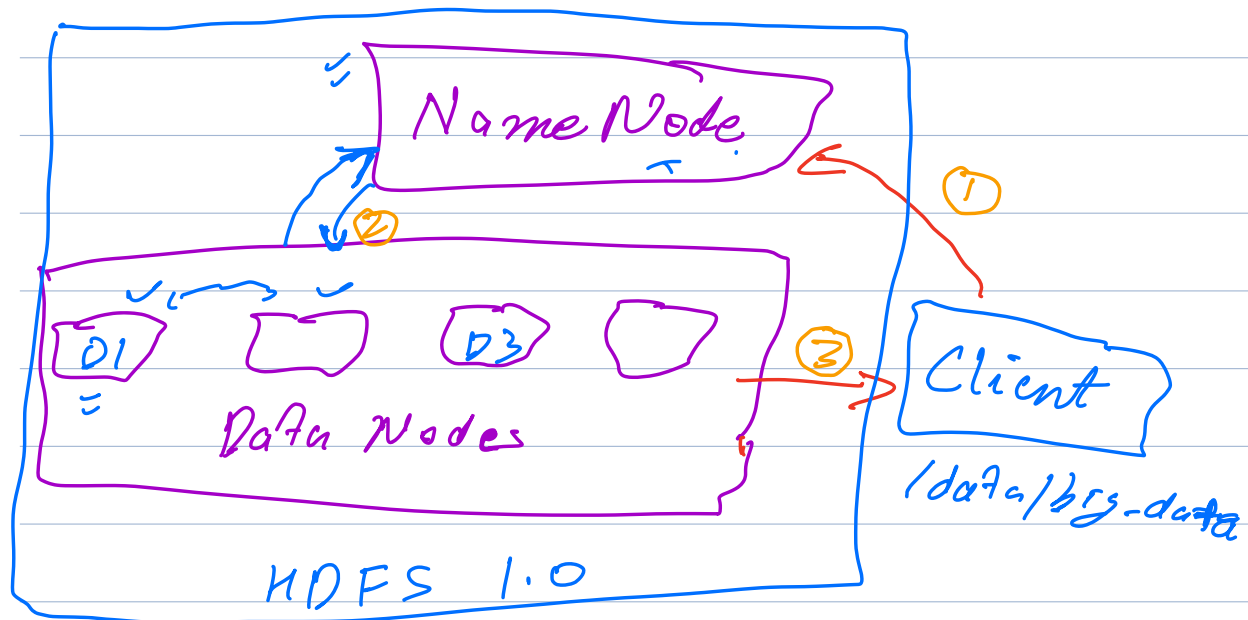
data/big-data -> [20] different parts
10 TB -> 500 GB each part

{data/big-data} part 1 => machine 1
part 2 => machine 2
⋮

store-data $\Rightarrow$ IS figure out how to
store $\Rightarrow$ maintain index

[Name Node] $\Rightarrow$ index

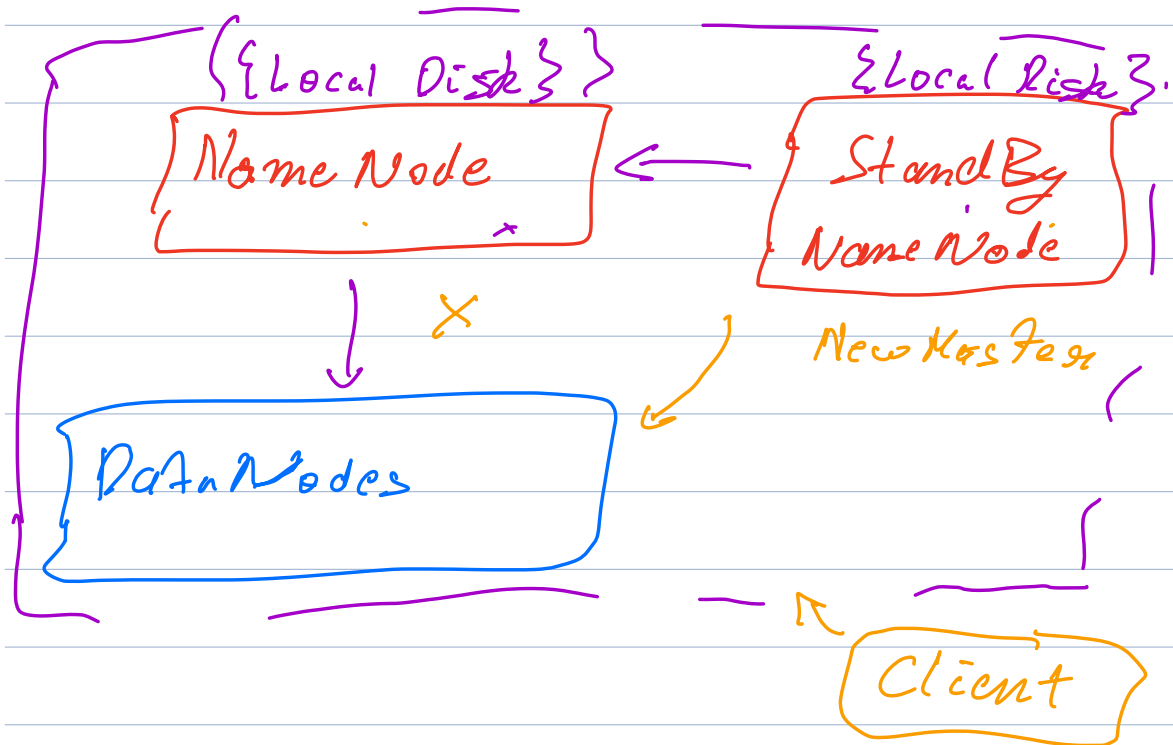
[Data Node] stores data



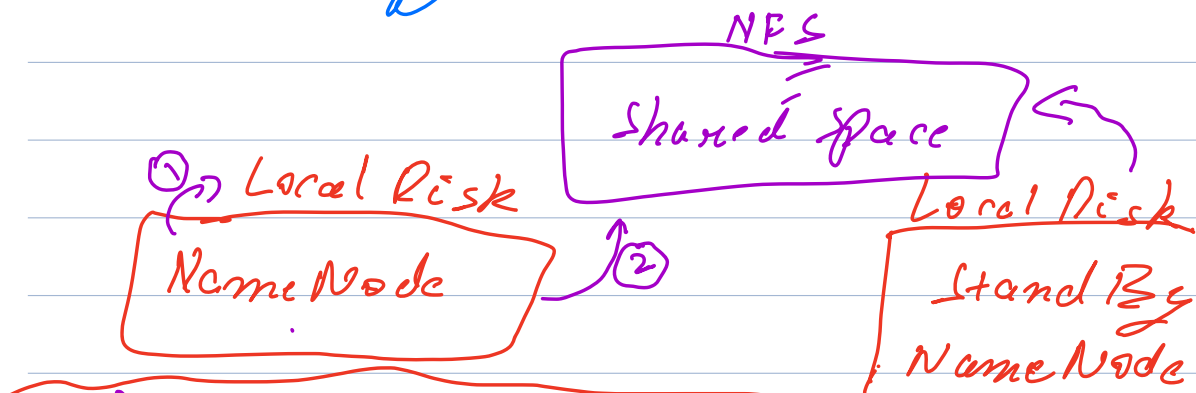
Hadoop 2.0



High Availability $\Rightarrow$ requirement



$\Rightarrow$ NameNode dies & StandBy was not able to copy



Heartbeat $< 3 \text{ sec}$

Heartbeat is much low $[0.5 \text{ sec}]$

→ Flooding

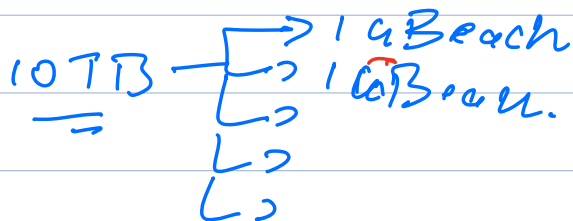
→ Node is slow not down

Heartbeat too high $[10 \text{ sec}]$

→ Downtime unnoticed

1 data / big data $\rightarrow [10 \text{ TB}]$

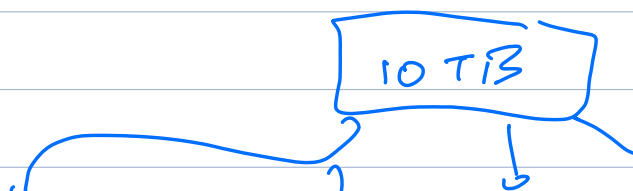
each machine is of 1 TB $[333 \text{ GB}]$



Replication factor $\Rightarrow$
 $\underline{3}$

10240 chunks $\Rightarrow [1024]$

machines

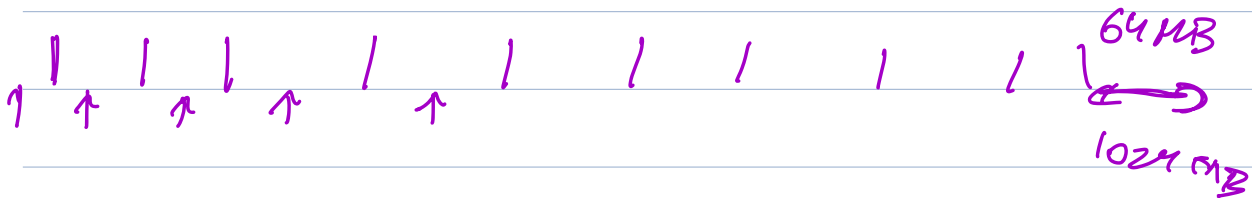
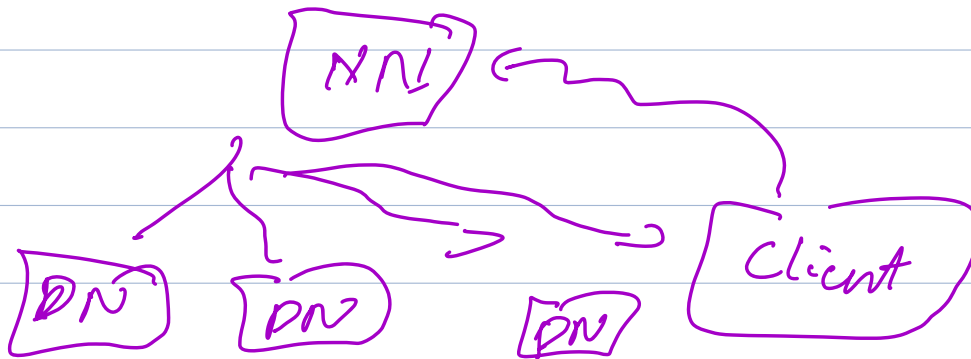


$\Rightarrow$ 1 machine $\Rightarrow$ 10 chunks

1GB
↓

1GB 1GB

500MB $\Rightarrow$ reduce chunk size $\Rightarrow$ count $\uparrow$



[Hdfs-site.xml] $\Rightarrow$ configuration file of HDFS

① dfs.replication $\Rightarrow$ 3

② dfs.namenode.dir $\Rightarrow$ namenode dir.

③ dfs.datanode.data.dir = datanode dir

0

④ dfs.block.size $\Rightarrow$ chunk size $\Rightarrow$ 128MB

⑤ dfs.namenode.http-address

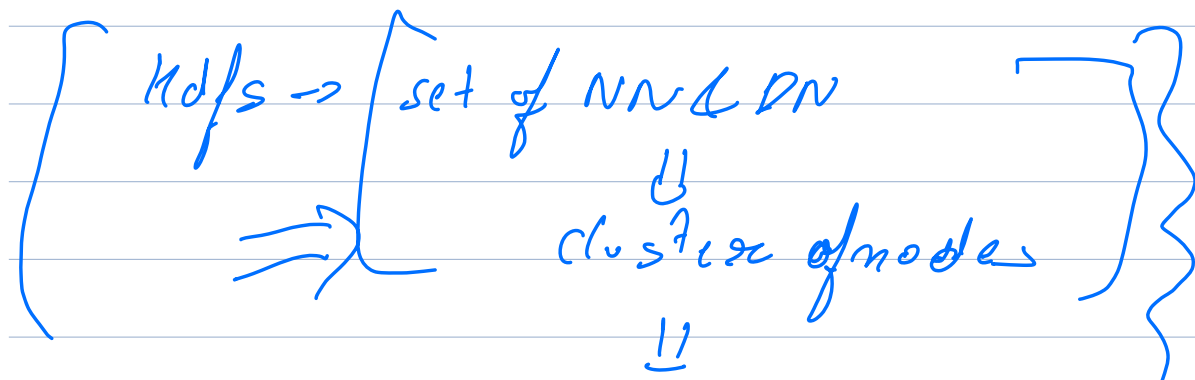
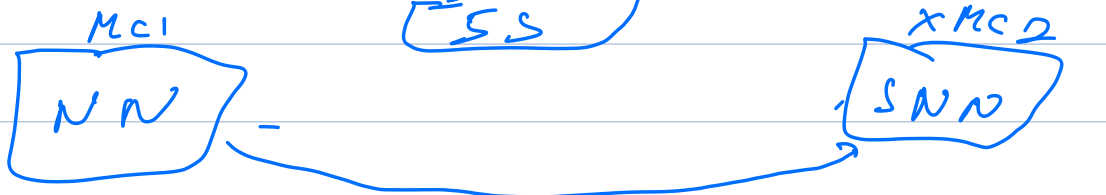
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Doubt session



nf $\Rightarrow$ 3 $\Rightarrow$ 2

0 =

NN $\Rightarrow$ $\left(\begin{array}{c} \text{root} \\ \vdots \\ \text{nimash.k} \end{array} \right)$

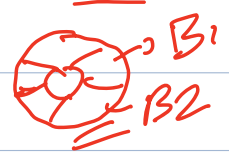
[P1]
DN = stores the data

Structure

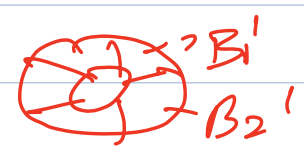
500GB
1TB

Client $\rightarrow$ hfs dfs - mkdir /root/user

NN $\rightarrow$ /root/user $\rightarrow$ $\begin{cases} 500GB \rightarrow P1 \rightarrow D1 \\ 500GB \rightarrow P2 \rightarrow D2 \end{cases}$



DN1
 $\Rightarrow$



DN2

NN $\Rightarrow$ root/user/part $\rightarrow$ D1 [B1, B2]